

Team Predators

Problem statement 01: AI for Digital Public Infrastructure & Governance

Our solution: INFRA-AI

A multilingual, location-first infrastructure intelligence platform for India, built as a Digital Public Good.



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INFRA-AI

Pin any place in India. See what was built, what is coming, and what people are asking for — every claim cited.

1,42,905 km

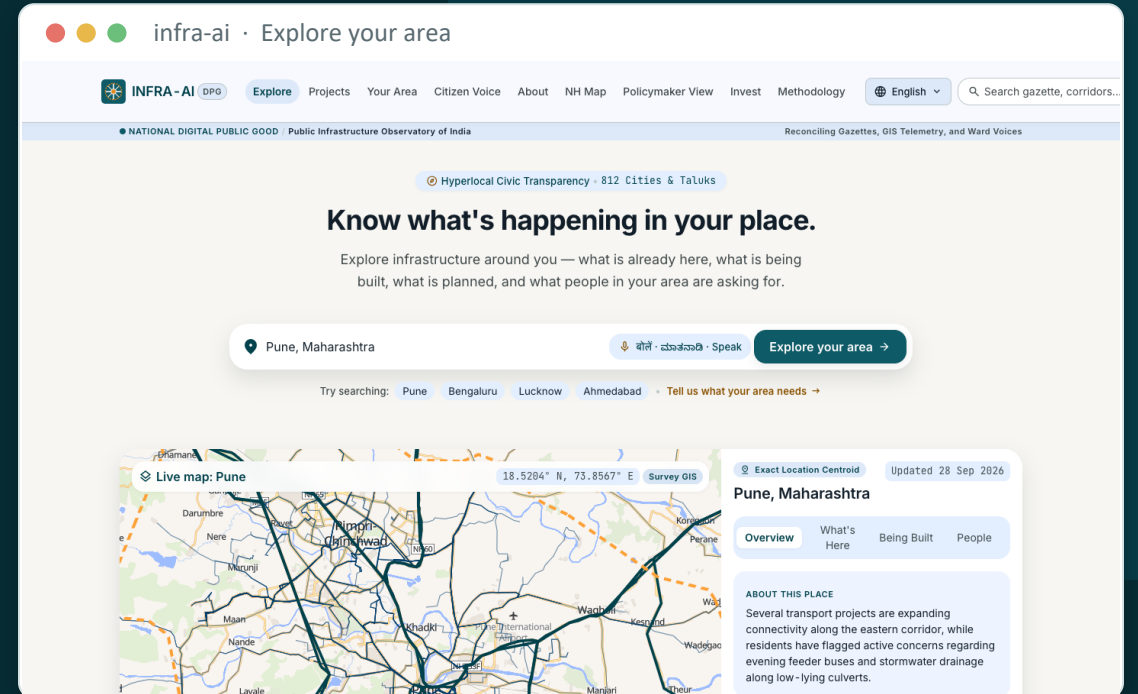
national highways mapped

381

cities scored

22

languages targeted



Problem statement 01

AI for Digital Public Infrastructure & Governance · Build with AI: Code for Communities, Second Edition

THE PROBLEM

Governments across India struggle to consolidate citizen feedback and align it with national infrastructure priorities. Development requests live in fragmented systems, leading to misaligned public spending, unaddressed infrastructure gaps, and no way to measure the impact of large-scale digital public infrastructure initiatives.

THE CHALLENGE

Build a scalable, multilingual AI platform, designed as a Digital Public Good, that aggregates citizen development requests via voice, text, and messaging apps across diverse linguistic regions of India. The system should analyse large datasets combining citizen feedback with national demographic data, infrastructure indices, and public investment plans, surfacing demand hotspots and recommending high-priority development projects to national policymakers.

How INFRA-AI answers it

Every request, every language

Voice, text and messaging intake in 22 languages

One view of a place

Requests joined to demographics, infrastructure and investment plans

Hotspots and priorities

Explainable scores for policymakers, each with cited facts

Open by design

A Digital Public Good: open data, open code, privacy by default

Infrastructure data is scattered — so needs go unseen

Citizen requests, plans and spending live in separate systems and languages. Nobody sees one place whole.



Map apps & OSM

Show what exists. Miss history, plans, project status.



Agency PDFs

Each agency shows its own projects. No common map.



Tender portals

Earliest signal of new work. No plain-language meaning, no location.



News & releases

Announcements and delays. No reliability check, no timeline.



Grievance portals

Single complaints. Never added up into demand.



Census & satellite

Who lives where. Not linked to any of the above.

What the gaps cost — national highways

73%

of NH length is two lanes or fewer

68%

of NHAI projects under construction are past schedule — median 19 months late

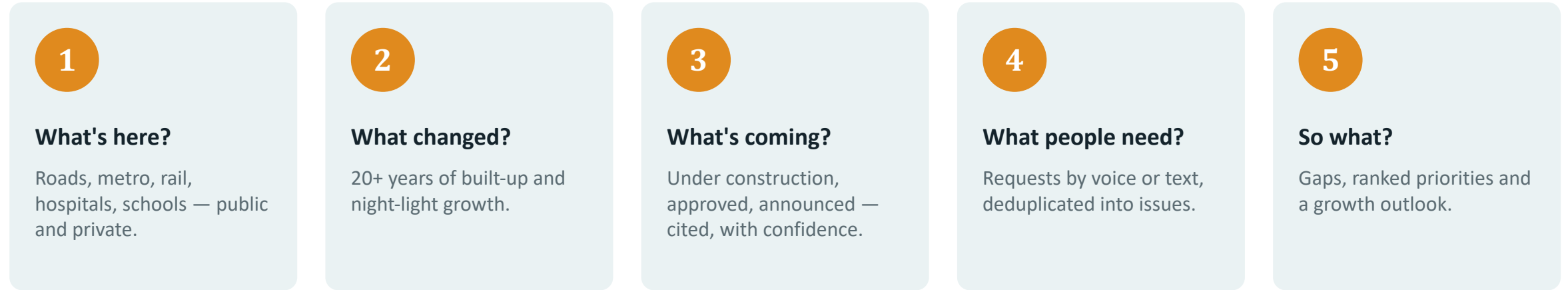
64,772

deaths on national highways in 2024, rising every year since 2020

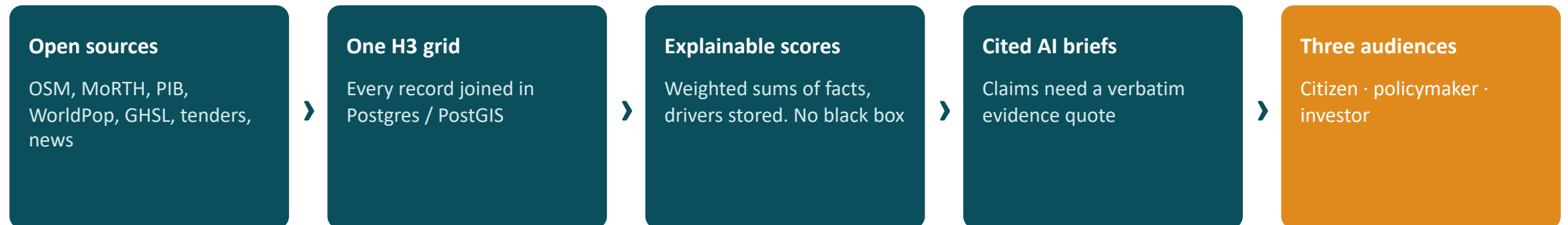
Sources: NHAI Datalake layer, MoRTH Road Accidents in India 2024; INFRA-AI analysis as of 2026-09-29.

One pin answers five questions

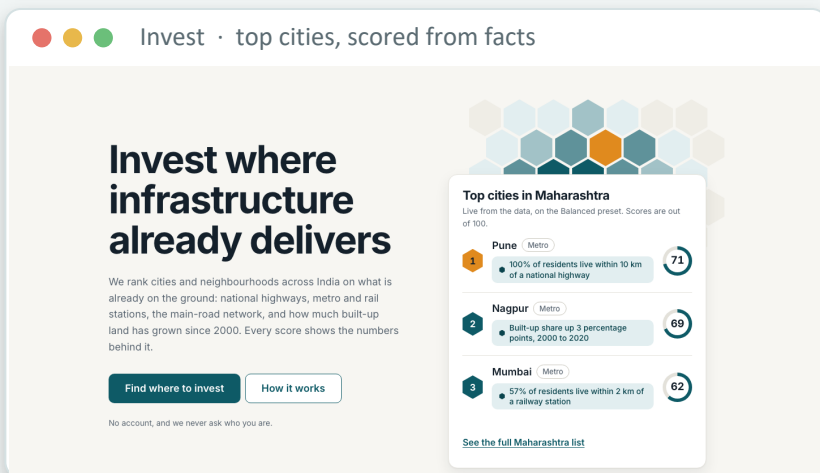
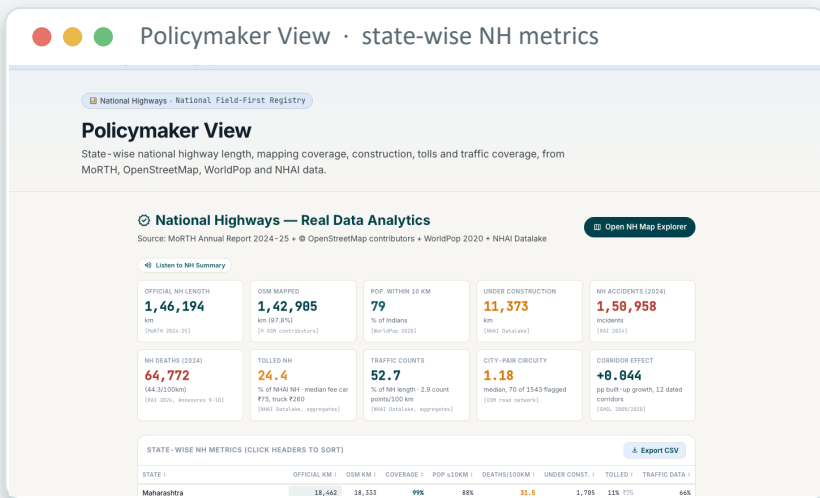
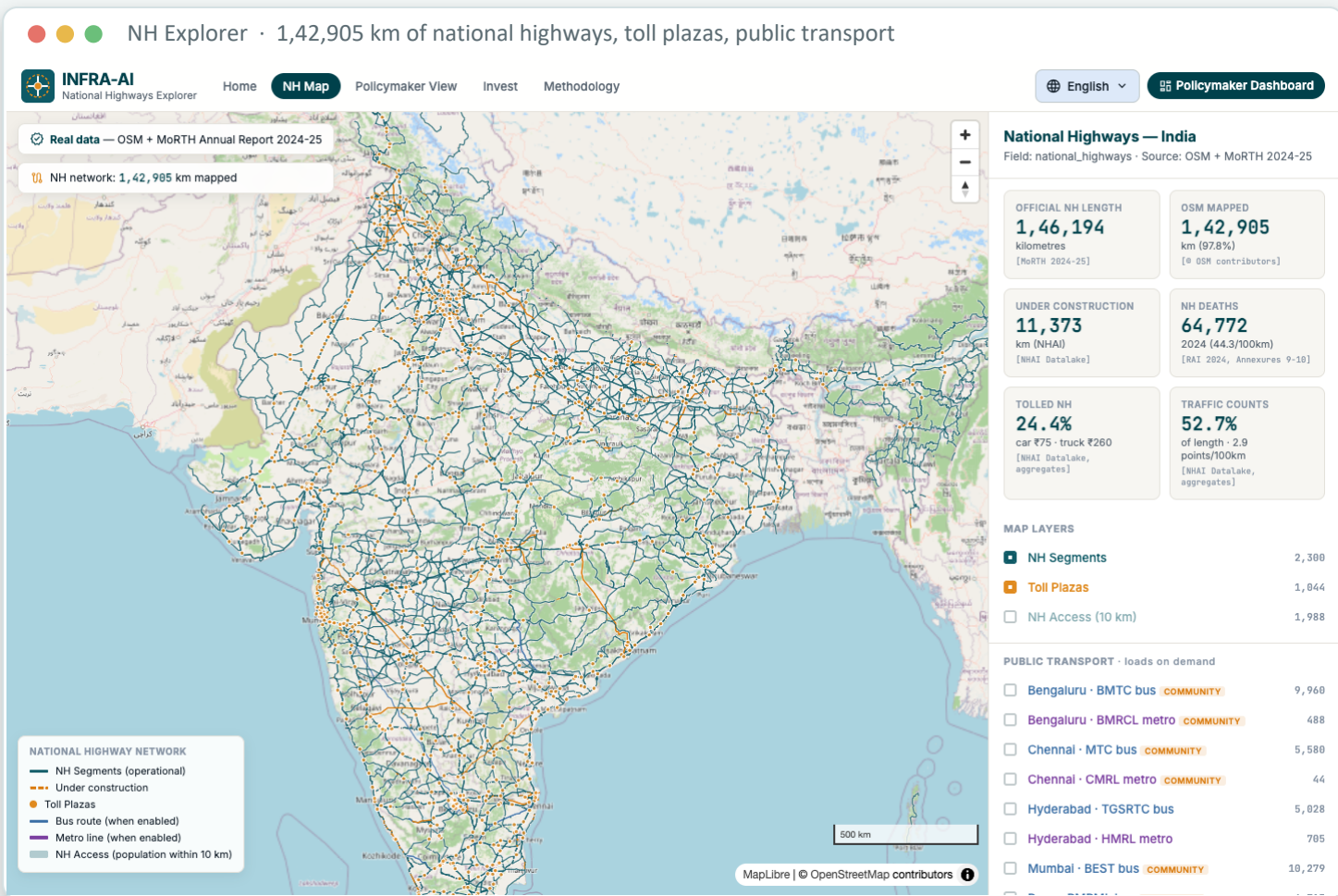
Join every dataset by place and time — then let people in any language ask, and policymakers act.



HOW IT WORKS



Working today — on real data, not mockups



Screenshots of the running app (fixture data as of 2026-09-29). Every number links to a source.

Feasible: data exists, stack is lean, core is built

Data

Proven

Open, licensed sources: OSM, MoRTH, WorldPop, GHSL, PIB. OSM covers 97.8% of official NH length (1,42,905 of 1,46,194 km). 381 cities scored.

Technology

Built, DB next

Postgres/PostGIS on an H3 grid, Go API, Python for ML. Scores are config-weighted sums with stored drivers — no black-box model.

Legal & privacy

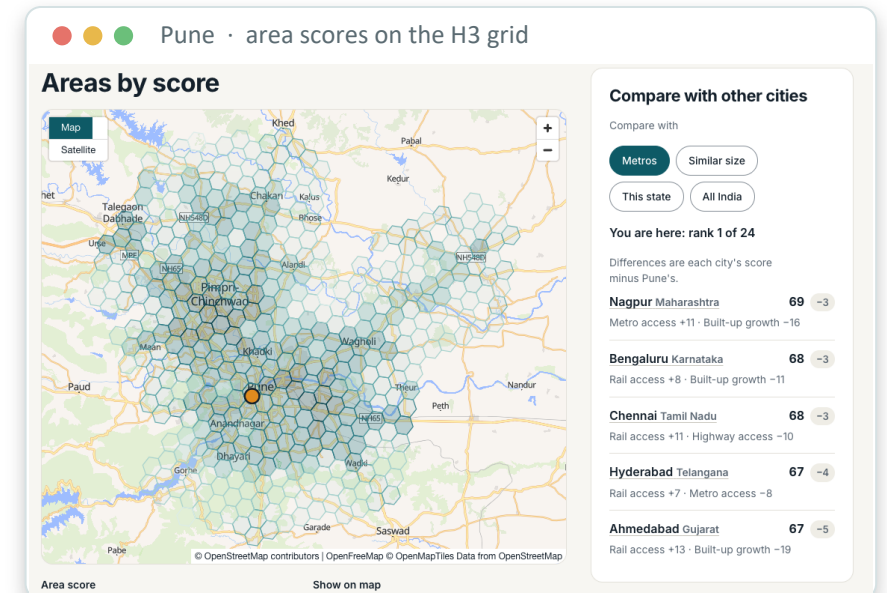
By design

Redact before storage; counts under 5 suppressed; never collect Aadhaar. Every record carries source, licence and confidence.

Delivery

On track

NH data and analysis done; pipeline half done. Investor flow live. Next: DB, API and UI shell. A new city is one config file.



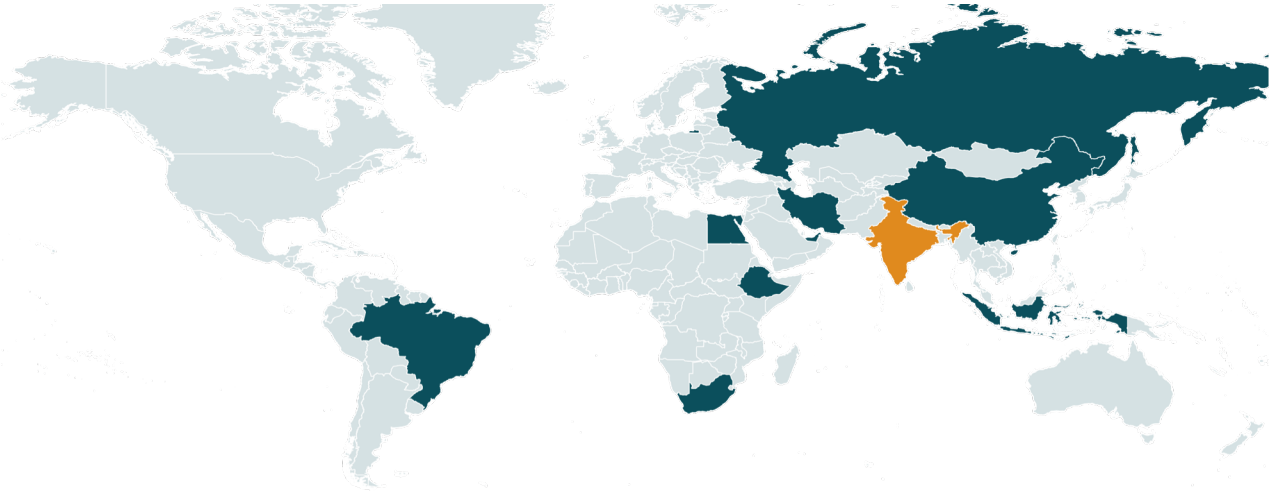
Risks we track

- NHA layer has no stated licence: used for analysis, never redistributed
- OSM lags openings: checked against official dates
- Ambiguous or invalid input goes to a review queue, never guessed

BRICS: one infrastructure lens

PROPOSED · NOT BUILT

Today INFRA-AI covers India only. Here is what the same platform could do for the ten BRICS members.



● India: live today
 ● Other members: proposed
 ● Not in scope

MEMBERS AND MAIN LANGUAGES

India	Hindi, English +20	Brazil	Portuguese
Russia	Russian	China	Mandarin
South Africa	English +10	Indonesia	Bahasa
Egypt	Arabic	Ethiopia	Amharic
Iran	Persian	UAE	Arabic

● **Why it can travel**

- A new place is one config file
- Global open data: OSM, WorldPop, GHSL
- Open-source DPG; UI already has pt, ru, zh

● **What we can do**

- Pilot one city per member
- Comparable access scores across countries
- Voice and text intake in national languages

● **What we should do first**

- Write an ADR: scope is India only today
- Licences, boundaries, privacy law per country
- Gold-set quality check per language

Borders: Natural Earth, India view; not the official Survey of India layer. Members as of 2025. Scores describe infrastructure, never rank countries. Nothing here is built yet.